

Unemployment and Inflation

ECON 101 - Macroeconomics

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- 5.1 Measuring the Unemployment Rate and the Labour Force Participation Rate
- 5.2 Types of Unemployment
- 5.3 Explaining Unemployment
- 5.4 Measuring Inflation
- 5.5 Using Price Indexes to Adjust for the Effects of Inflation
- 5.6 Real versus Nominal Interest Rates
- 5.7 Does Inflation Impose Costs on the Economy?

Measuring Unemployment and Inflation

- Previous chapter: how to measure **total output** (GDP).
- This chapter: how to measure
 - **Unemployment** in the labour market, and
 - **Inflation** in the goods and services market.
- Both are central for:
 - macroeconomic policy,
 - evaluating economic performance,
 - everyday decisions of households and firms.

5.1 Learning Objective

Goal

Understand how unemployment, labour force participation, and employment rates are measured.

- Canada is a large economy – millions of people, changing jobs all the time.
- We cannot track every person individually in real time.
- Instead, Statistics Canada uses **survey data** to estimate:
 - employment,
 - unemployment,
 - labour force participation,
 - employment-population ratio.

The Labour Force and Unemployment

- **Labour force:** sum of **employed** and **unemployed** workers in the economy.
- **Unemployment rate:** percentage of the labour force that is unemployed:

$$\text{Unemployment rate} = \frac{\text{Number of unemployed}}{\text{Labour force}} \times 100.$$

- These are monthly headline numbers often reported in the news.

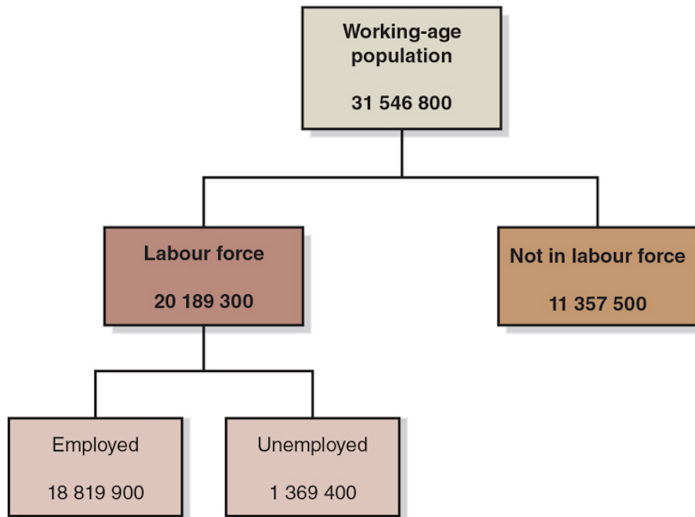
The Labour Force Survey (LFS)

- Each month, Statistics Canada conducts the **Labour Force Survey (LFS)**.
- Roughly **56,000 households** are selected to be representative.
- Members of **working age** (15 years and older) are:
 - asked about employment status, and
 - asked about recent **job search** activities.
- Based on the answers, each person is classified as:
 - **Employed**
 - **Unemployed**
 - **Not in the labour force**

Definitions: Employed, Unemployed, Not in the Labour Force

- **Employed:**
 - Did paid work, or
 - Did unpaid work for a family business, or
 - Worked for themselves, or
 - Were temporarily away from their jobs (vacation, illness, strike).
- **Unemployed:**
 - Not currently at work,
 - *Available* for work,
 - Actively looked for work in the previous four weeks.
- **Not in the labour force:**
 - Neither employed nor unemployed (students not seeking work, retirees, etc.).

Figure 5.1 – Employment Status of the Working-Age Population



- Labour force participation rate:

$$\text{LFPR} = \frac{\text{Labour force}}{\text{Working-age population}} \times 100.$$

- Measures what fraction of working-age people are **actively** in the labour market.
- Helps us distinguish:
 - low unemployment because people have jobs,
 - vs. low unemployment because many have given up looking for work.

- Employment-population ratio:

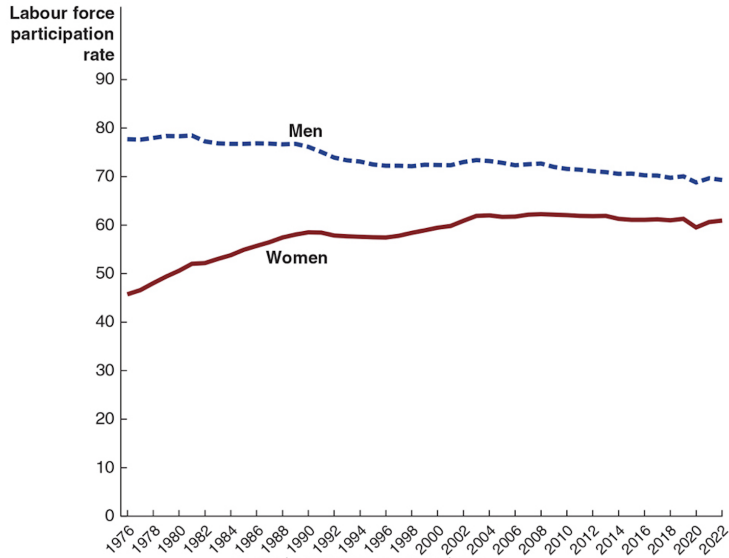
$$\text{Employment-population ratio} = \frac{\text{Number employed}}{\text{Working-age population}} \times 100.$$

- This tells us how many people **actually have jobs**, relative to the working-age population.
- Useful indicator of the **health of the labour market** during expansions and recessions.

Problems with Measuring the Unemployment Rate

- The measured unemployment rate is **not perfect**.
- It may **understate** true unemployment:
 - Distinguishing between **unemployed** and **not in the labour force** can be difficult.
 - **Discouraged workers**: people who have stopped looking for work but would take a job if offered one.
 - It does not measure intensity of employment — e.g., part-time vs. full-time, underemployment.
- It may **overstate** unemployment:
 - Some people may claim to be unemployed while not seriously looking, to keep benefits or avoid taxes.

Figure 5.2 – Trends in Labour Force Participation



- The labour market is **dynamic**:
 - Firms constantly create and destroy jobs.
- Example (from the slide):
 - In January 2022, there were 890,500 more people employed than in January 2021.
 - But the number of jobs **created** was much larger, because many jobs were also **destroyed**.
- Month to month, some provinces can see large job losses even while the national economy is improving.

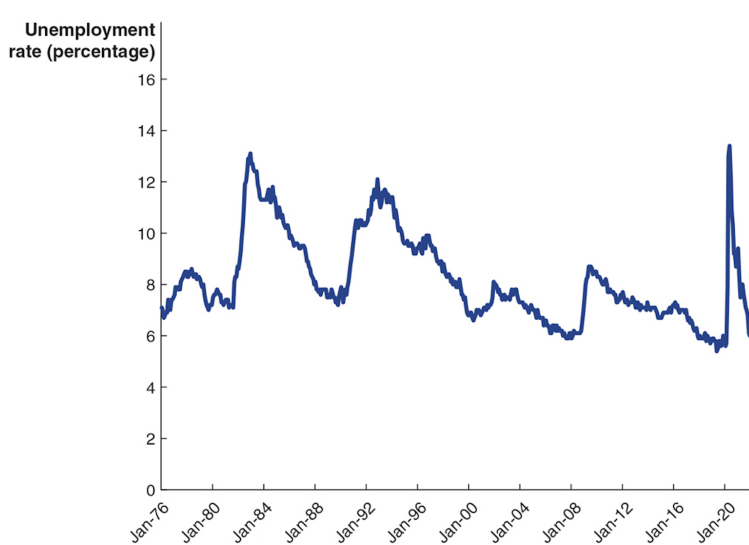
5.2 Learning Objective

Goal

Identify and understand the four types of unemployment.

- Unemployment is not a single phenomenon.
- To interpret unemployment correctly, we distinguish:
 - frictional,
 - structural,
 - cyclical,
 - seasonal unemployment.

Figure 5.3 – The Unemployment Rate in Canada



Four Types of Unemployment

- Frictional unemployment
- Structural unemployment
- Cyclical unemployment
- Seasonal unemployment
- Each type has different causes and different policy implications.

- **Frictional unemployment:**
 - Short-term unemployment that arises from the process of matching workers with jobs.
- Mainly due to **job search**:
 - People entering or re-entering the labour force.
 - People between jobs.
- Some frictional unemployment is **unavoidable**:
 - People need time to learn about opportunities and find good matches.
- In fact, some frictional unemployment can **increase** economic efficiency:
 - It allows workers to find jobs that better fit their skills and preferences.

Example: Frictional Unemployment

- A recent university graduate spends three months searching for a job in their field.
- They are counted as **unemployed**, but this is frictional unemployment.
- From the economy's perspective:
 - It is better for them to search a bit longer and find a suitable, productive job than to accept the first random job.

- **Structural unemployment:**
 - Unemployment that arises from a persistent **mismatch** between the skills and attributes of workers and the requirements of jobs.
- Often associated with:
 - technological change,
 - changes in demand for certain industries,
 - regional shifts in economic activity.
- Structural unemployment tends to be **longer-term:**
 - Workers may need retraining or relocation.

Example: Structural Unemployment

- A manufacturing worker loses their job due to automation.
- Their skills do not easily transfer to growing sectors such as IT or health care.
- Without retraining, they may remain unemployed for a long period.

- **Cyclical unemployment:**
 - Unemployment caused by a **business cycle recession**.
- When aggregate demand falls:
 - firms reduce output,
 - and lay off workers even if workers are skilled and jobs are otherwise a good match.
- Example:
 - In February 2009, Chrysler temporarily closed its assembly plant in Brampton, Ontario, due to low sales.
 - Laid-off employees experienced **cyclical unemployment**.

- **Seasonal unemployment:**
 - Unemployment due to seasonal factors such as weather or calendar patterns.
- Examples:
 - Construction and tourism often slow in winter.
 - Agriculture, fisheries, and some retail sectors are highly seasonal.
- Seasonal unemployment can make the measured unemployment rate:
 - artificially high in winter,
 - artificially low in summer.
- To address this, Statistics Canada provides:
 - **seasonally adjusted** series (seasonal effects removed),
 - **unadjusted** series (seasonal effects included).

Full Employment and the Natural Rate of Unemployment

- Even in good times, there is always some unemployment.
- When all unemployment is due to frictional and structural factors, the economy is at **full employment**.
- **Natural rate of unemployment:**
 - The normal rate of unemployment, consisting of frictional and structural unemployment.
- Estimates for Canada suggest a natural rate around **5.6–6.7%** in recent years.

5.3 Learning Objective

Goal

Explain what determines the unemployment rate and how policy affects it.

- Governments often try to influence unemployment directly and indirectly.
- Some policies **reduce** frictional or structural unemployment.
- Other policies can unintentionally **increase** unemployment.

- Examples of policies that can **reduce unemployment**:
 - Retraining programs (e.g., Ontario's "Better Jobs Ontario") to reduce structural unemployment.
 - Hiring subsidies or job search assistance to reduce frictional unemployment.
- Examples of policies that might **increase unemployment**:
 - Employment insurance (if very generous),
 - Minimum wage laws that raise the cost of hiring,
 - Some labour market regulations.

Employment Insurance (EI)

- Suppose you lose your job. You can:
 - take a new low-paying job immediately, or
 - search for a better job that matches your skills.
- If **employment insurance** payments are available, you are more likely to choose the second option.
- In Canada, EI replaces about **55%** of earnings up to a weekly maximum (e.g., \$595 per week in 2021).
- EI effects:
 - Allows more time for job search $\Rightarrow$ better matches, higher productivity.
 - But also tends to **increase** the average duration of unemployment.

Minimum Wage Laws

- Minimum wage laws are designed to help low-income workers.
- Raising the minimum wage increases the **cost of labour**.
- Firms may respond by:
 - hiring fewer workers,
 - substituting capital for labour,
 - reducing hours.
- In Canada, minimum wages vary by province:

Region	Example Minimum Wage (June 2022)
Nunavut	\$16.00/hour
New Brunswick	\$12.75/hour

- Studies suggest a 10% increase in the minimum wage might reduce teenage employment by about 2%.
- Overall effect on total unemployment is believed to be **relatively small**.

- **Labour unions:** organizations of workers that bargain with employers for:
 - higher wages,
 - better working conditions.
- Unions can push wages above the competitive level for their members.
- In Canada:
 - about **30.9%** of workers belonged to a union by the end of 2021,
 - roughly **77.2%** of government employees are unionized,
 - only about **15.3%** of private-sector workers are unionized.
- Conclusion: unions likely have **limited impact** on overall unemployment in Canada.

- **Efficiency wage:** a wage above the market-clearing level that firms pay to increase workers' productivity.
- Why pay more than necessary?
 - Higher wages can reduce worker turnover.
 - They can increase effort and reduce shirking.
 - They can attract higher-quality applicants.
- But paying an efficiency wage means:
 - firms hire fewer workers,
 - some workers who want jobs at that wage cannot find them.
- This contributes to **persistent unemployment**, even when the economy is not in recession.

5.4 Learning Objective

Goal

Define the price level and inflation rate, and understand how they are computed.

- We introduced the **price level** in the previous chapter using the GDP deflator.
- Now we look at more detailed measures used in practice:
 - the **Consumer Price Index (CPI)**,
 - the **Producer Price Index (PPI)**.

- **Price level:** a measure of the average prices of goods and services in the economy.
- **Inflation rate:** the percentage increase in the price level from one year to the next:

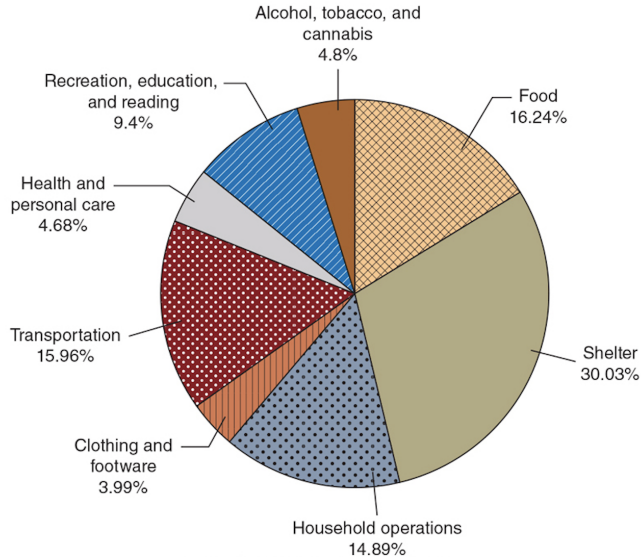
$$\text{Inflation rate} = \frac{\text{Price level}_t - \text{Price level}_{t-1}}{\text{Price level}_{t-1}} \times 100.$$

- In Chapter 4, we used the **GDP deflator** to measure the price level for all final goods and services.
- However, households and firms are often more interested in specific baskets of goods.

The Consumer Price Index (CPI)

- **CPI:** measures the average change over time in the prices paid by a typical household for the goods and services they purchase.
- It is based on a fixed **basket of goods and services**.
- Example categories:
 - food, shelter, transportation, clothing, recreation, etc.
- CPI is widely used to:
 - index wages,
 - adjust pensions and benefits,
 - compare the cost of living over time.

Figure 5.4 – The CPI Basket



Calculating the CPI

- To compute the CPI for a given year, we need:
 - A fixed **basket of goods and services** (quantities from a base year).
 - The cost of buying the basket in the **base year**.
 - The cost of buying the basket in the **current year**.

- Then:

$$CPI_t = \frac{\text{Cost of basket in year } t}{\text{Cost of basket in base year}} \times 100.$$

- By convention, $CPI_{\text{base year}} = 100$.

A Simple CPI Calculation (1 of 2)

- Suppose the CPI basket (from 2002) has:
 - 10 units of good A,
 - 5 units of good B,
 - 2 units of good C.
- We know prices for 2002, 2022, and 2023 from the slide table.
- **Cost of basket in 2002:**

$$\text{Cost}_{2002} = 10P_A^{2002} + 5P_B^{2002} + 2P_C^{2002}.$$

A Simple CPI Calculation (2 of 2)

- Cost of basket in 2022:

$$\text{Cost}_{2022} = 10P_A^{2022} + 5P_B^{2022} + 2P_C^{2022}.$$

- CPI in 2022:

$$\text{CPI}_{2022} = \frac{\text{Cost}_{2022}}{\text{Cost}_{2002}} \times 100.$$

- Similarly for 2023.
- Inflation between 2022 and 2023:

$$\text{Inflation} = \frac{\text{CPI}_{2023} - \text{CPI}_{2022}}{\text{CPI}_{2022}} \times 100.$$

- This inflation rate is often called **CPI inflation** or a **cost-of-living** measure.

Apply the Concept: Were Prices Really Cheaper in the Past?

- Example from the slides:
 - In 1915, a pair of cashmere socks cost \$0.35.
 - In 2021, a similar pair costs \$39.00.
- Nominally, socks are much more expensive today.
- But incomes are also much higher.
- Using the CPI to convert \$0.35 (1915) into 2021 dollars gives about \$8.13.
- Compared with the actual 2021 price of \$39, socks are more expensive in real terms, but relative to **average income**, households today can afford many more socks than in the past.

Is the CPI Accurate?

- The CPI is widely used, but it is not perfect.
- Four main biases:
 - **Substitution bias**
 - **Increase in quality bias**
 - **New product bias**
 - **Outlet bias**
- Together, these biases tend to make CPI inflation **overstate** the true increase in the cost of living (by about 0.5 to 1 percentage point per year).

- **Substitution bias:**
 - Basket is fixed; in reality, consumers substitute toward relatively cheaper goods.
- **Quality change bias:**
 - Part of the price increase reflects better quality (e.g., safer, more efficient cars).
- **New product bias:**
 - New goods (e.g., smartphones, tablets) enter the basket with a delay.
- **Outlet bias:**
 - Shift from traditional stores to discount outlets or online retailers is not fully captured by the fixed basket.

Producer Price Index (PPI)

- **PPI:** an average of the prices received by producers of goods and services at all stages of production.
- Uses a basket of:
 - raw materials,
 - intermediate goods,
 - goods sold by wholesalers.
- Changes in the PPI often signal future movements in consumer prices:
 - higher input costs today tend to lead to higher retail prices later.

5.5 Learning Objective

Goal

Use price indexes to adjust nominal values for inflation.

- Often we want to compare **purchasing power** across years.
- To do this, we convert nominal amounts into **real** amounts using a price index like the CPI.

Adjusting Nominal Values for Inflation

- Suppose your mother earned \$30,000 in 1993.
- \$30,000 in 1993 could buy more goods and services than \$30,000 in 2021.
- To express the 1993 salary in 2021 dollars:

$$\text{Value in 2021 dollars} = \text{Value in 1993 dollars} \times \frac{\text{CPI}_{2021}}{\text{CPI}_{1993}}.$$

- Using the actual CPI numbers from the slides, this yields about \$49,626 for 2021.

- Unions and employers often negotiate **cost-of-living adjustments (COLAs)**.
- Government programs (pensions, social benefits) are often indexed to CPI.
- This helps protect real purchasing power when prices rise.

5.6 Learning Objective

Goal

Distinguish between nominal and real interest rates.

- Interest rates are crucial for:
 - savers and borrowers,
 - investment decisions,
 - monetary policy.

- When you lend money, the borrower pays you back with interest.
- If the **nominal interest rate** is 6%, a \$1,000 loan repaid in one year returns \$1,060.
- But prices may rise over that year, reducing the **real purchasing power** of \$1,060.
- To understand the true cost of borrowing and lending, we use the **real interest rate**.

Nominal vs. Real Interest Rate

- **Nominal interest rate:** the stated interest rate on a loan.
- **Real interest rate** adjusts for inflation:

$$\text{Real interest rate} \approx \text{Nominal interest rate} - \text{Inflation rate.}$$

- Example:
 - If nominal rate is 6% and inflation is 2%, the real rate is about 4%.
- For low to moderate inflation, this approximation is very accurate and widely used in macroeconomics.

5.7 Learning Objective

Goal

Discuss the problems inflation can cause, even when anticipated.

- At first glance, inflation might seem harmless if all prices and wages doubled together.
- But in reality, inflation creates **winners and losers** and can distort decisions.

Is Inflation a Problem?

- If all prices and wages rose at exactly the same rate:
 - relative prices would not change,
 - everyone's purchasing power would stay the same.
- In practice:
 - not all prices adjust at the same time or by the same amount,
 - some wages are fixed in long-term contracts,
 - some assets (like cash) lose value.
- So inflation affects:
 - the **distribution of income and wealth**,
 - the incentives to save, borrow, and invest.

Problems with Anticipated Inflation

- Even if inflation is **anticipated**, it can create costs:
 - **Redistribution of income:**
 - Some workers' wages adjust fully to inflation; others lag behind.
 - **Loss of purchasing power of cash:**
 - People holding currency see its real value fall.
 - **Menu costs:**
 - Firms incur costs of changing prices (new catalogues, re-tagging, updating systems).
 - **Bracket creep and taxes on nominal returns:**
 - Investors are taxed on nominal, not real, returns; inflation increases the tax burden even if real returns are unchanged.

Problems with Unanticipated Inflation

- **Unanticipated inflation** is especially damaging.
- It makes it hard to write long-term contracts.
- Borrowers and lenders are exposed to risk:
 - If inflation turns out higher than expected:
 - borrowers gain (repay in less valuable dollars),
 - lenders lose (receive lower real interest).
 - If inflation turns out lower than expected:
 - borrowers lose,
 - lenders gain.
- Unpredictable inflation discourages saving and long-term investment.

Example: High and Uncertain Inflation

- In periods of high and unstable inflation (e.g., late 1970s–early 1980s):
 - Mortgage interest rates climbed into double digits (18% or more).
 - Banks tried to protect themselves from high and uncertain inflation.
- Households who locked in at high nominal rates suffered later when inflation fell but their nominal rates did not.
- These experiences show why central banks place high value on **low and stable inflation**.

- The unemployment rate, labour force participation rate, and employment-population ratio summarize labour market conditions.
- Unemployment has four main types: frictional, structural, cyclical, and seasonal.
- Policies such as EI, minimum wage laws, unions, and efficiency wages affect unemployment in different ways.
- Inflation is measured using price indexes like the CPI, PPI, and GDP deflator.
- The CPI is a cost-of-living index but contains several biases.
- We use price indexes to convert nominal values into real values.
- Real interest rates matter for saving, borrowing, and investment decisions.
- Inflation, especially when unanticipated, can impose significant costs on the economy.

Thank you!